INTRODUCTION TO FLUID MECHANICS

Department of Chemical Engineering

CHEM ENG 2O04
Course Notes

Instrumentation

Fox & McDonald

10th & 9th editions: 3.1 (pressure); 8.9 (flow rate)

Vlachopoulos book - Chapter 23

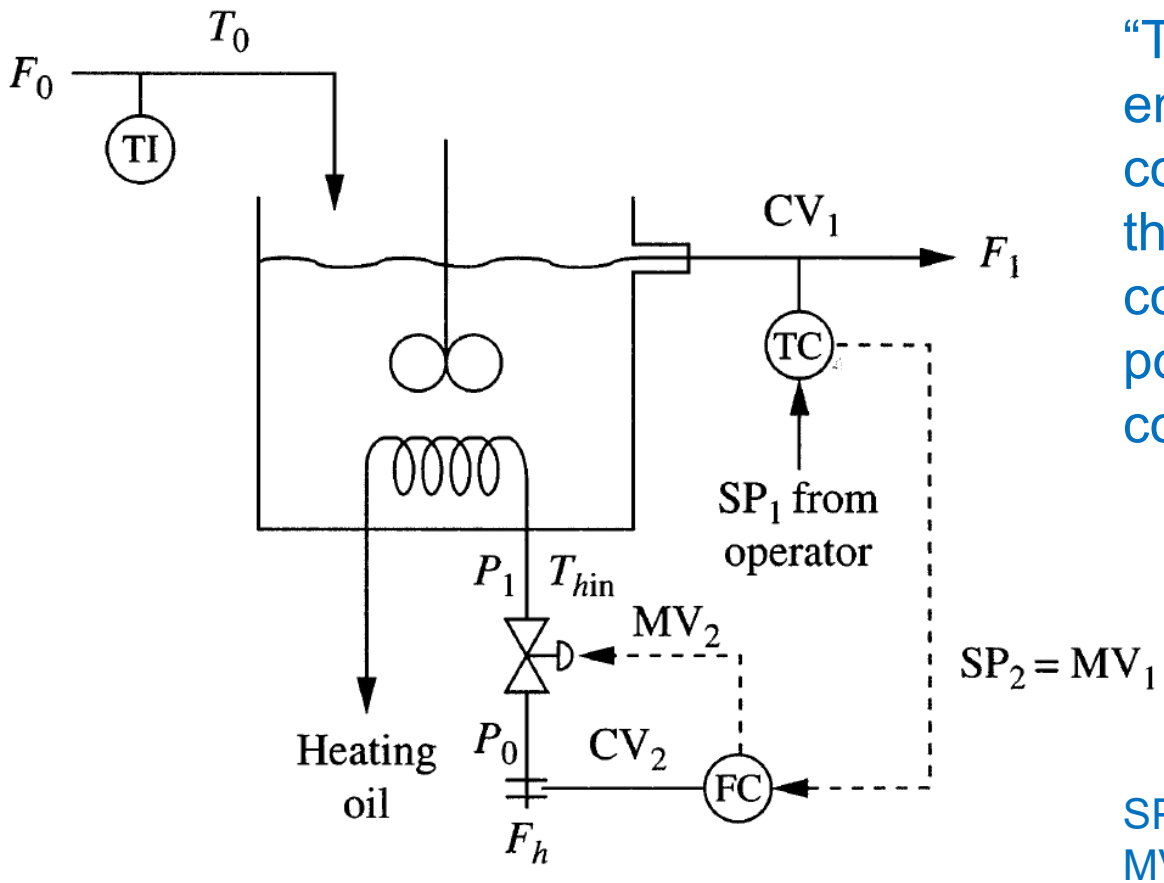
"Piping System Fundamentals – Chapter 6 - Valves"

"The function of education is to teach one to think intensively and to think critically. Intelligence plus character – that is the goal of true education"

Dr. Martin Luther King Jr.

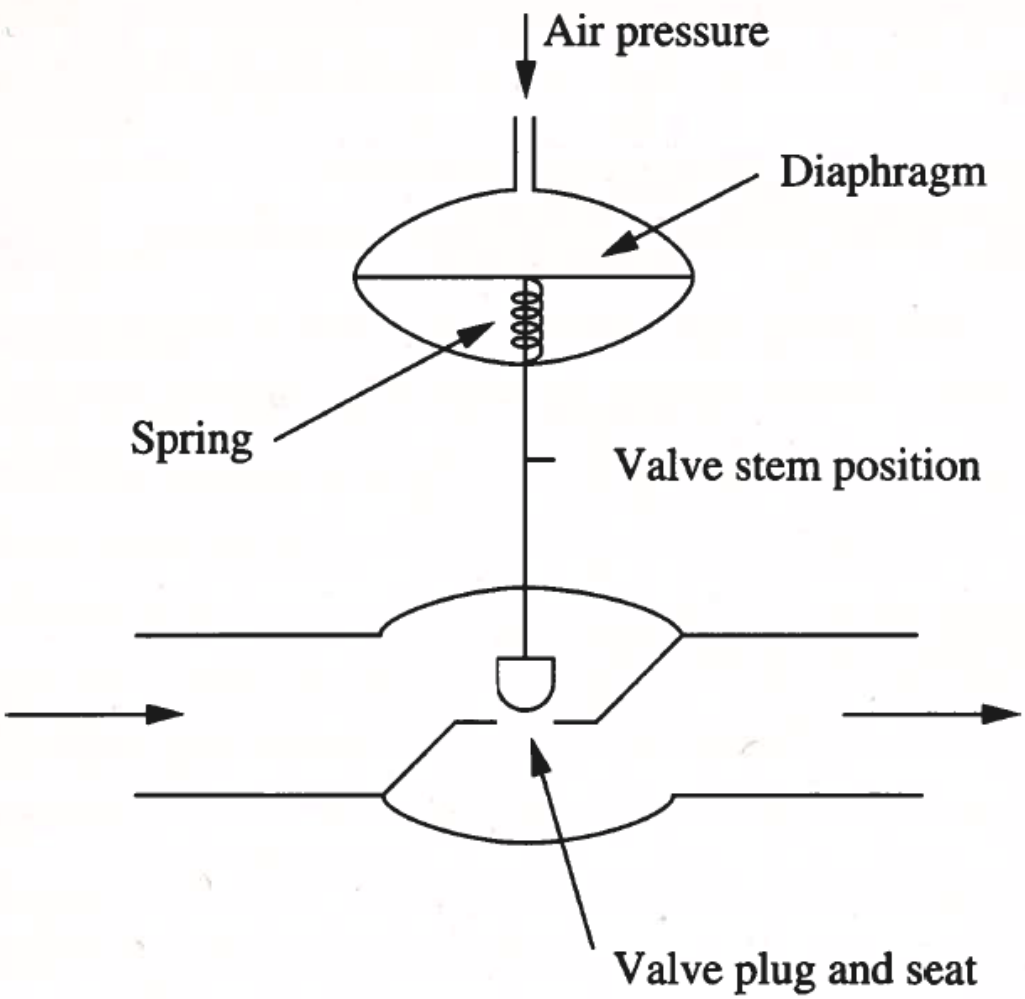
Motivation (cont'd)

- From ChE 3P04:
  - A stirred-tank heat exchanger with <u>cascade control</u>



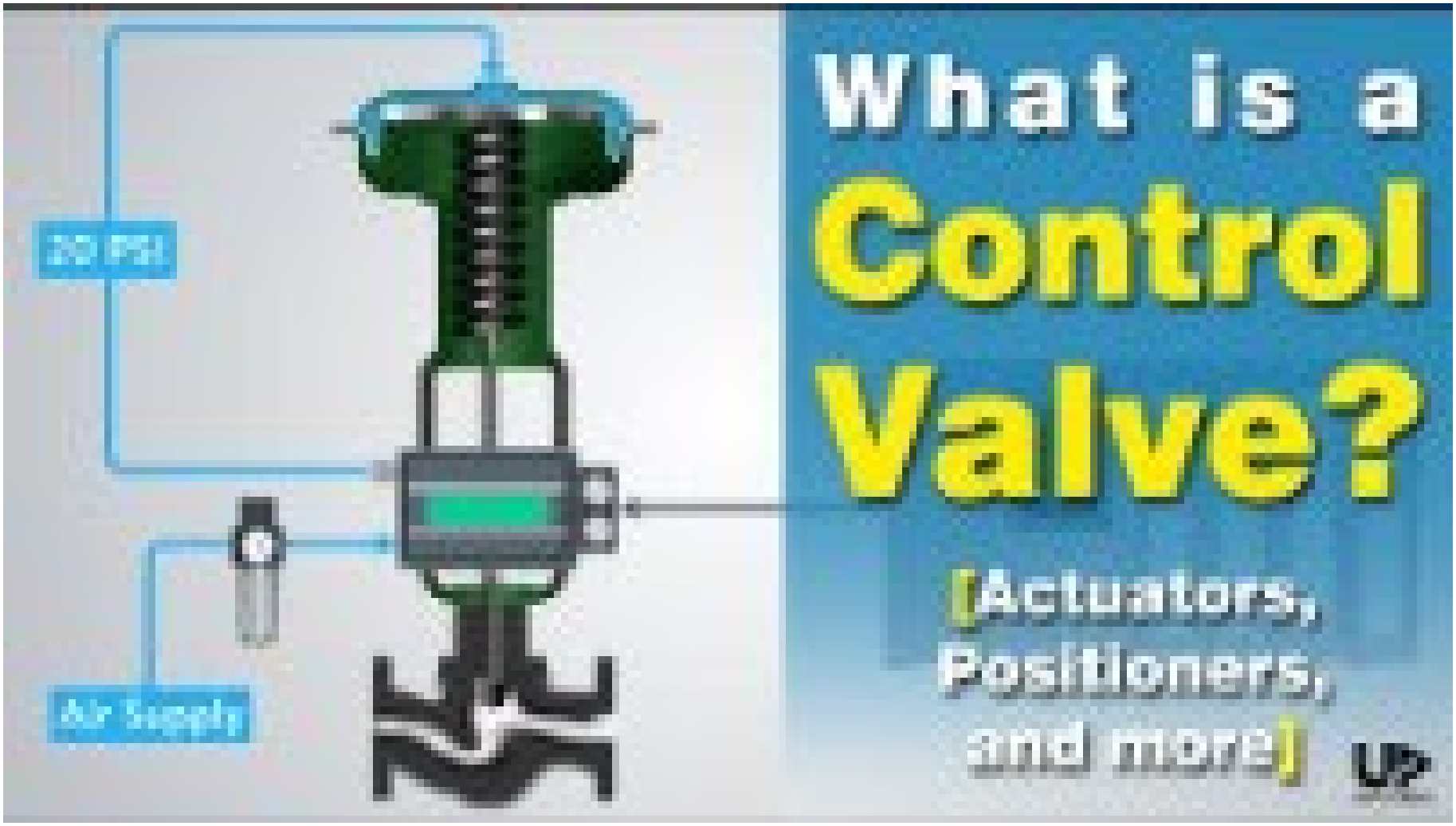
"The control system employs two feedback controllers….the output of the exit temperature controller adjusts the set point of the flow controller"

SP ≡ set point
MV ≡ manipulated variable
CV ≡ controlled variable



From 3P04 textbook:
"The process fluid flows through the opening in the valve, with the amount open (or resistance to flow) determined by the valve stem position. The valve stem is connected to the diaphragm (a flexible metal sheet that can bend in response to forces). The two forces acting on the diaphragm are the spring and the variable pressure from the control signal."

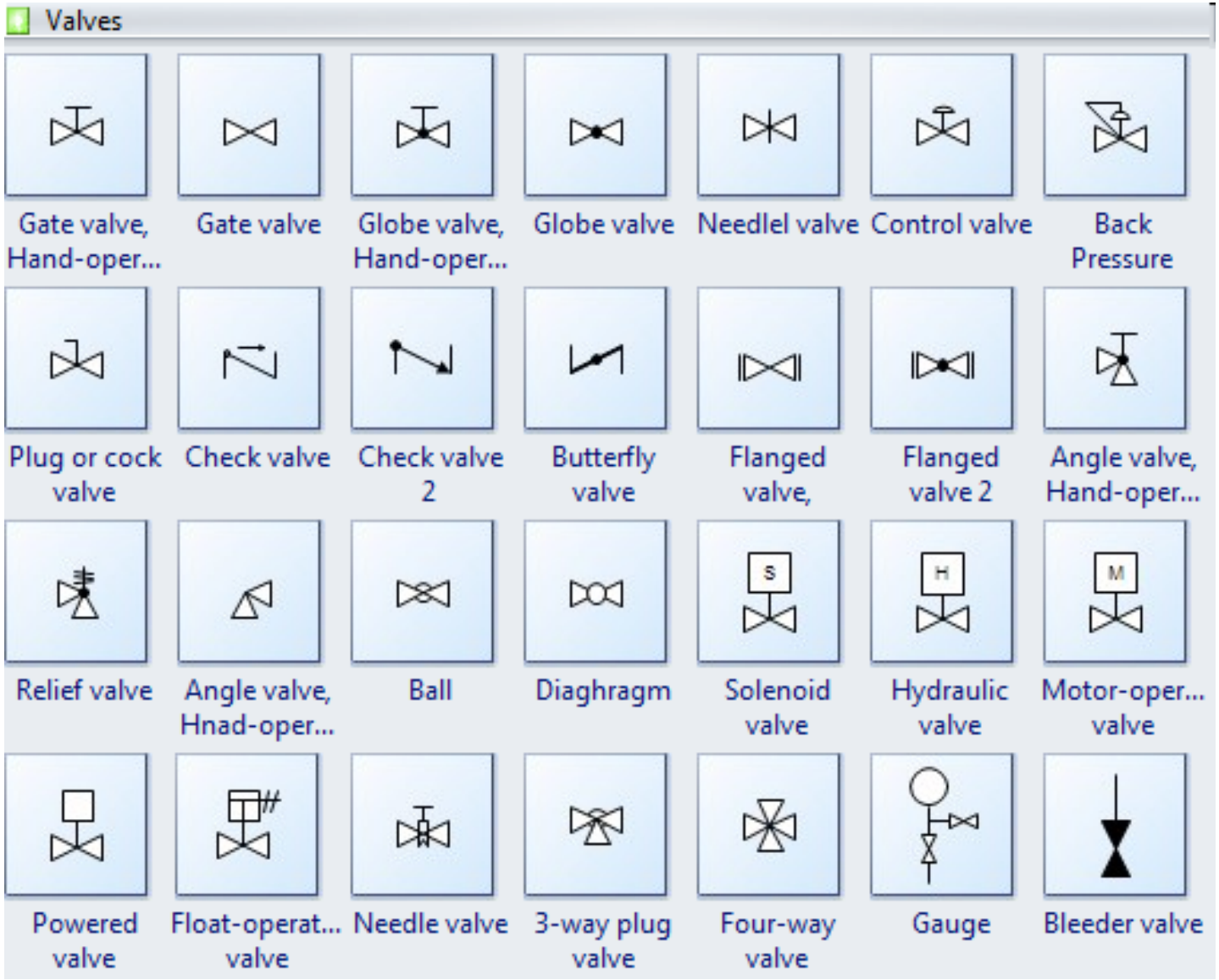
FIGURE 7.2
Schematic of control valve.


What is a
Control
Valve?
(Actuators,
Positioners,
and more)
20 PSI
Air Supply

The World's Largest Valve

Weighing at 117 tons and standing a majestic 40 feet tall, Blackhall's Zero-Leakage Metal Seated Parallel Faced Gate Valve has been designed to last for 100 years. The DN2800 / 108 inch dia Gate valves are designed and manufactured for America's 1st platinum rated water pipeline project IPL (Integrated Pipeline). The project will drastically boost the water supply to Dallas-Fort Worth Metroplex by 1.59 billion Litres (350 Million Gallons) per day by 2060.

Piping & Instrumentation Diagram (P&ID) symbols

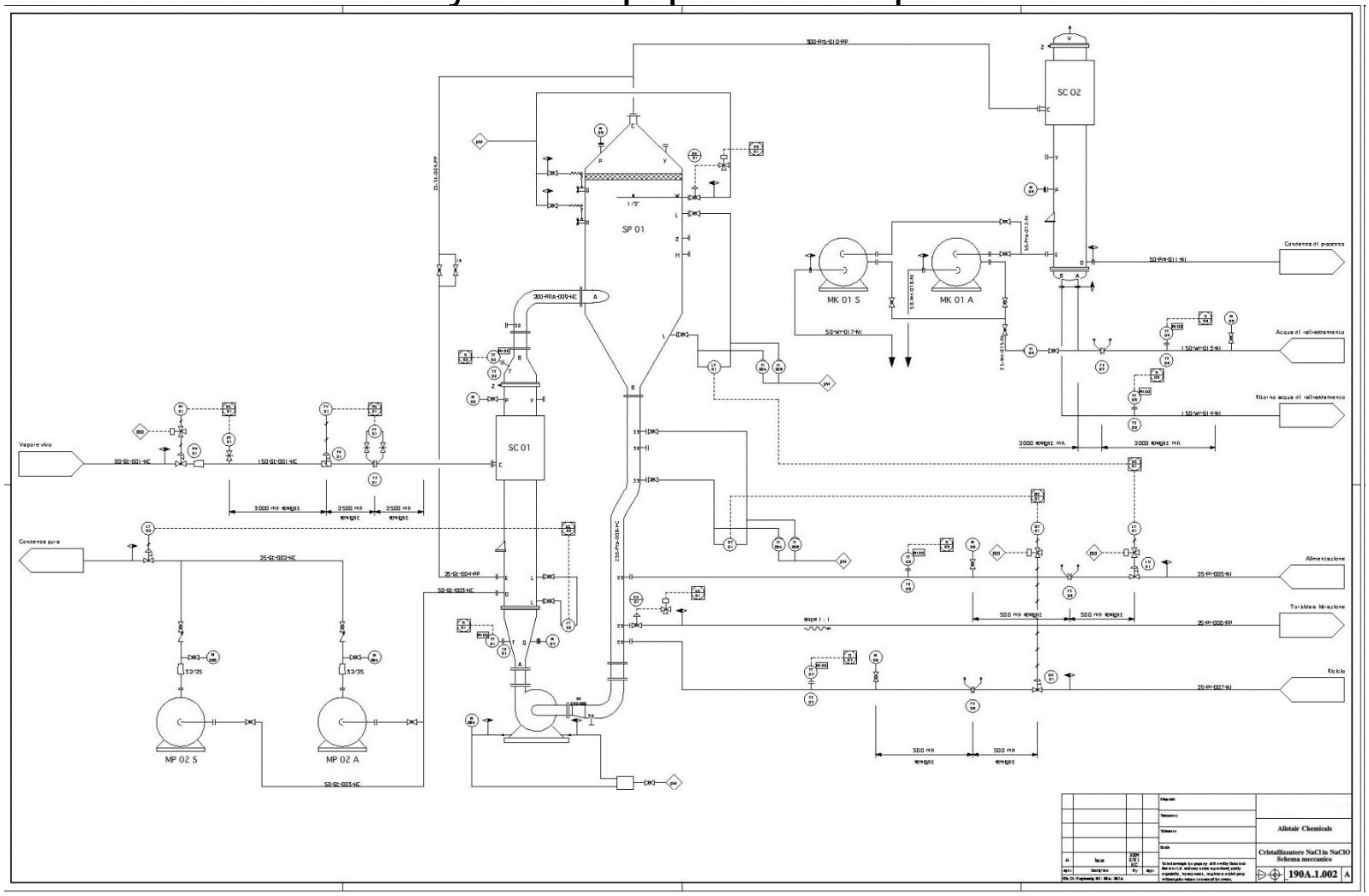


Read “Piping System Fundamentals – Chapter 6 - Valves” posted on AVENUE

- Isolation valves
  - Used to start and stop flow
  - Used to isolate a component or part of the piping system
  - Variety of types – the amount of resistance to flow when in the ‘fully open’ position varies with their design
    - Ball valve - lowest resistance to flow of any standard design
    - Gate valve
    - Plug valve
    - Butterfly valve
    - Globe valve
- Check valves
  - Prevent reversal of flow in a pipeline which could damage equipment
  - Variety of types:
    - Swing check valve
    - Tilting disc check valve
    - Lift check valve
    - Foot check valve
    - Stop check valve

Piping & Instrumentation Diagram (P&ID)

- “a schematic illustration of functional relationship of piping, instrumentation and system equipment components”



Piping & Instrumentation Diagram (P&ID)

- "a schematic illustration of functional relationship of piping, instrumentation and system equipment components"

A 'good' P&ID should include:

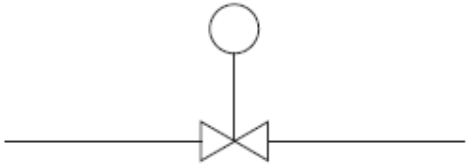
1. All process equipment identified by an equipment number
2. All pipes identified by a line number as well as the pipe size and material of construction.
3. All valves with an identification number; the type and size should be shown.
4. Ancillary fittings that are part of the piping system (sampling ports, strainers, etc.) with an identification number.
5. Pumps identified by a suitable code number.
6. All control loops and instruments, with an identification number.

A P&ID should NOT include:

•pressure temperature and flow data
•elbow, tees and similar standard fittings
•extensive explanatory notes

P&ID Symbols

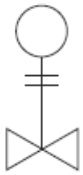
- Controls valve (both pneumatic and electric actuators):



- Failure mode:
  - The direction of the arrow shows the position of the valve on failure of the power supply

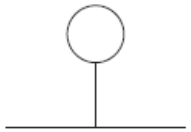
Fails open

Fails shut

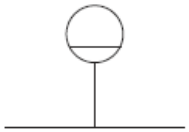


Maintains position

- Instruments and controllers:



Locally mounted



Main panel mounted

Controller & display are located out on the plant near to the sensing instrument location

Controller & display are located on a panel in the control room

Viscosity (aka 'dynamic viscosity' or 'absolute viscosity')

$$\tau = \mu \frac{dV}{dy} \quad \text{OR} \quad \mu = \frac{\tau}{\left(\frac{dV}{dy}\right)}$$

- Need independent measurements of shear stress (force divided by surface area) and shear rate (velocity gradient):
  - Rotational viscometer
  - Falling sphere viscometer

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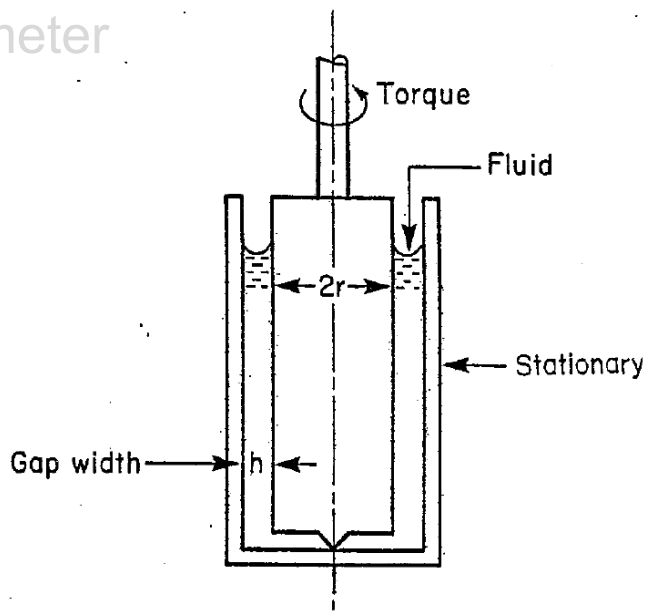

Fig. 1.4 Concentric cylinder viscometer. Torque and revolutions per minute (rpm) are directly measured.

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  - Rotational viscometer
  - Falling sphere viscometer

We will analyze how this instrument works in a tutorial

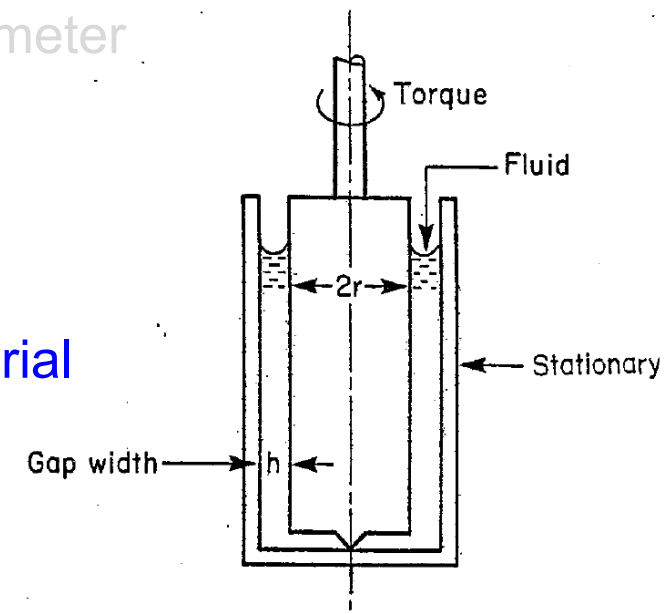


Fig. 1.4 Concentric cylinder viscometer. Torque and revolutions per minute (rpm) are directly measured.

Viscosity (aka 'dynamic viscosity' or 'absolute viscosity')

- Need independent measurements of shear stress (force divided by surface area) and shear rate (velocity gradient):
  - Rotational viscometer
  - Falling sphere viscometer:
    - Useful only for high viscosity fluids (so VERY slow movement!)
    - Measure time required for sphere (radius, R_S; density ρ_S) to travel a distance between fixed points in vertical column of fluid (density, ρ_f) → terminal velocity (V_T)

$$\mu = \frac{2{R_S}^2(\rho_s - \rho_f)g}{9V_T}$$

In-class problem

A falling sphere viscometer is used to calculate the viscosity of glycerin at room temperature. When a small marble (0.9 cm in diameter, specific gravity = 2.6) is dropped into a vertical column of glycerin, it takes 9.3 seconds to travel between the two marked positions that are 30 cm apart; the column has a diameter of 8 cm.

a. Draw a nice sketch of the system as described above

b. Calculate the viscosity (in Pa·s units) of glycerin.

c. Compare to reported value for glycerin of 1.49 Pa·s then comment on any reasons for the different in the calculated and reported values.

In-class problem

A falling sphe
room tempera
gravity = 2.6)
seconds to tra
the column ha

a. Draw a n
b. Calculate
c. Compare
any reas

Table A.2

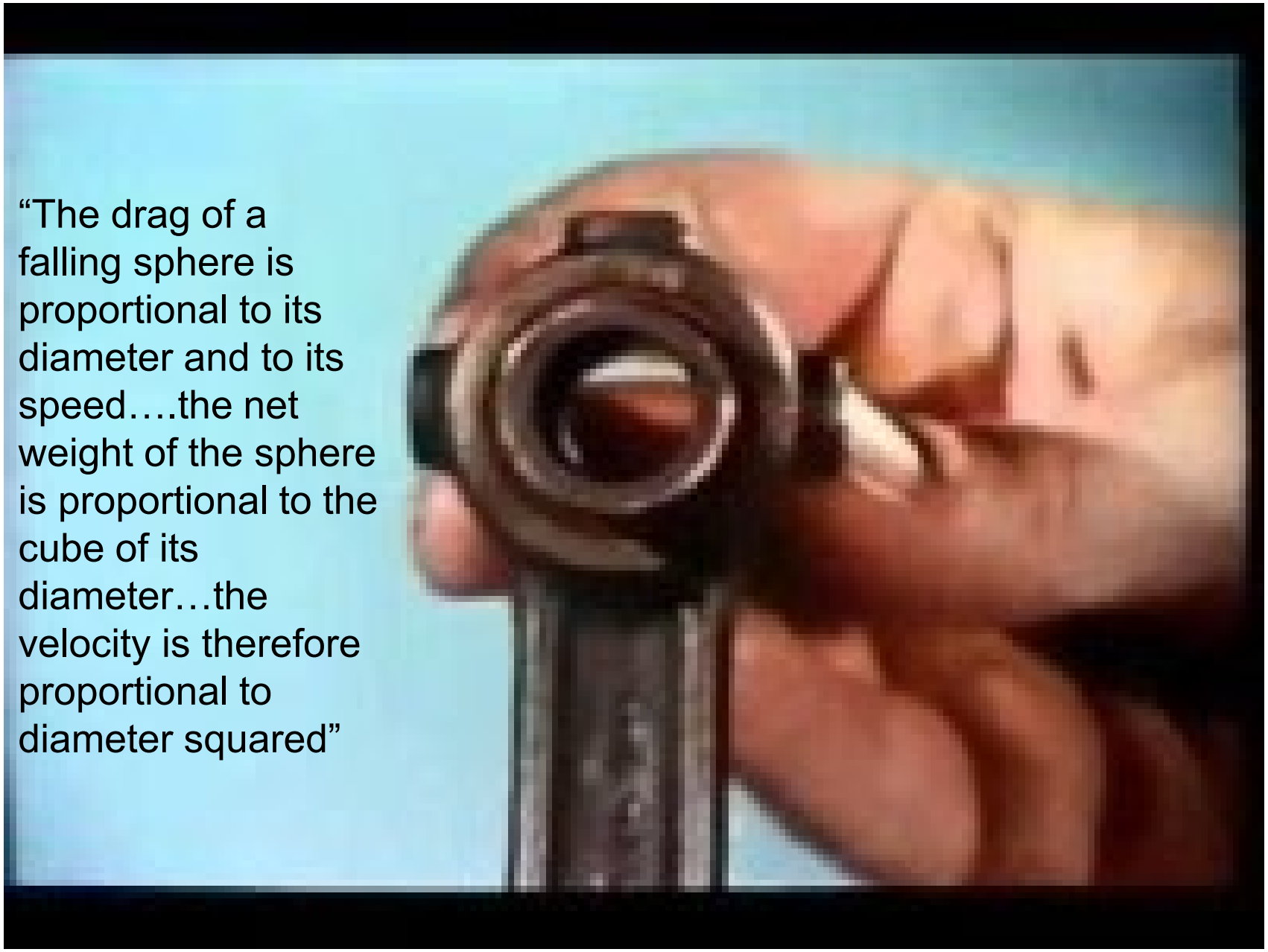
Physical Properties of Common Liquids at 20°C

Liquid	Isentropic Bulk Modulus[a] (GN/m^2)	Specific Gravity (—)
Benzene	1.48	0.879
Carbon tetrachloride	1.36	1.595
Castor oil	2.11	0.969
Crude oil	—	0.82–0.92
Ethanol	—	0.789
Gasoline	—	0.72
Glycerin	4.59	1.26
Heptane	0.886	0.684
Kerosene	1.43	0.82
Lubricating oil	1.44	0.88
Methanol	—	0.796
Mercury	28.5	13.55
Octane	0.963	0.702
Seawater[b]	2.42	1.025
SAE 10W oil	—	0.92
Water	2.24	0.998

Source: Data from References [1, 5, 6].

[a]Calculated from speed of sound; 1 GN/m^2 = 10^9 N/m^2 (1 N/m^2 = 1.45×10^{-4} lbf/in.^{2}).

[b]Dynamic viscosity of seawater at 20°C is $\mu = 1.08 \times 10^{-3}$ N · s/m^2. (Thus, the kinematic viscosity of seawater is about 5 percent higher than that of freshwater.)



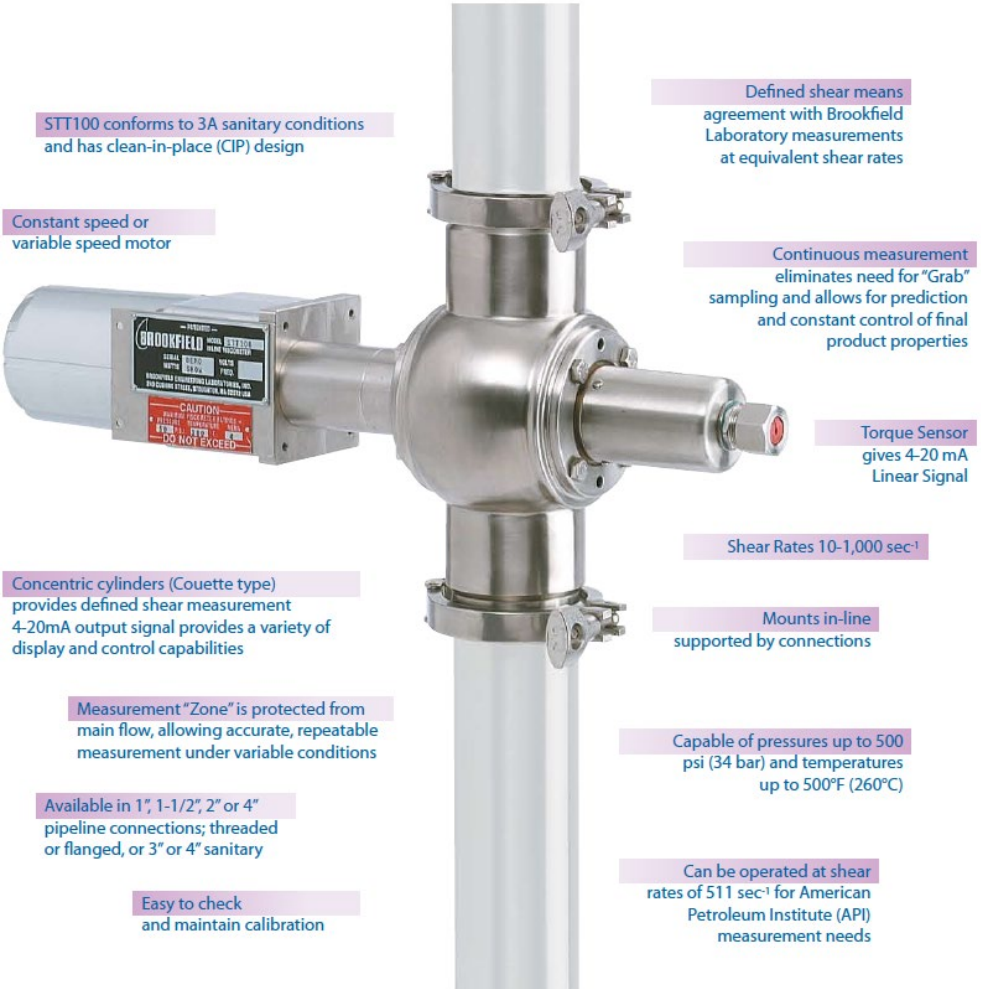
Link to Video (17:35 – 19:50)

Customized high-throughput rheometer (Anton Paar HTR 502)

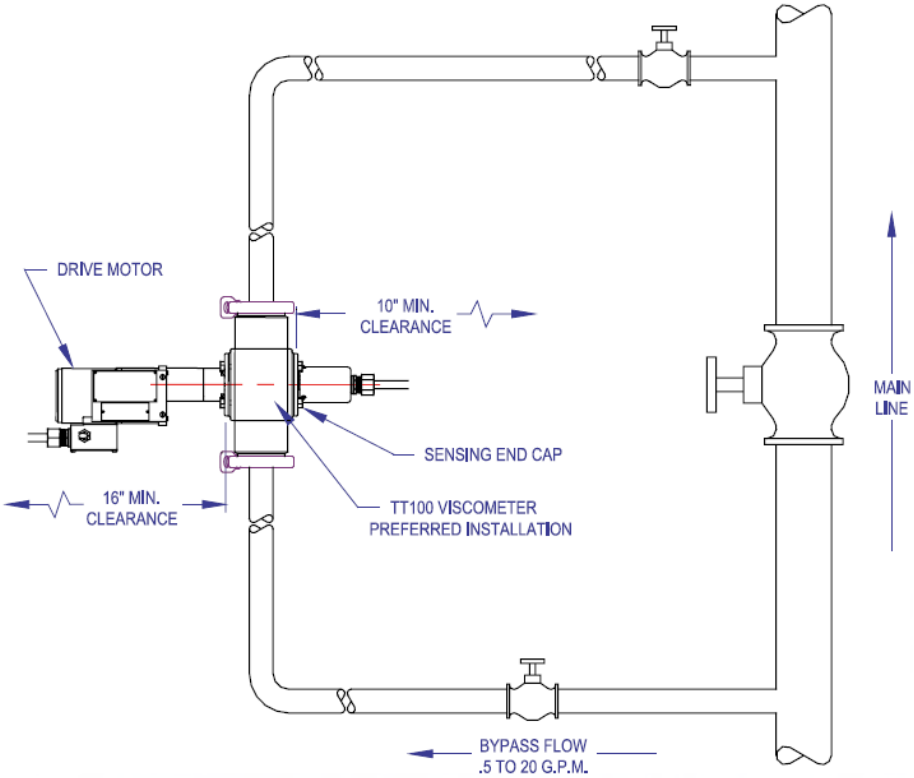
https://www.youtube.com/watch?v=_QCQ0aQzjG8&feature=youtu.be

In-line viscometer

- Brookfield STT-100



Typical Installation



Pressure Measurement

"pressure gauge" usually refers to a self-contained indicator that converts the detected process pressure into the mechanical motion of a pointer

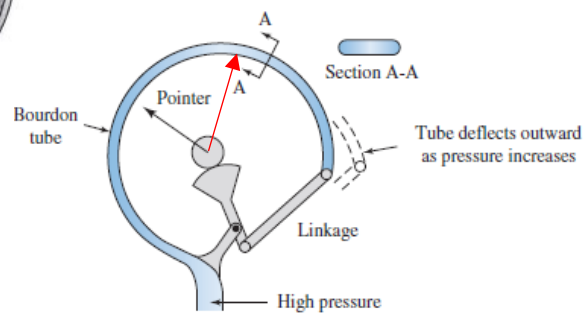


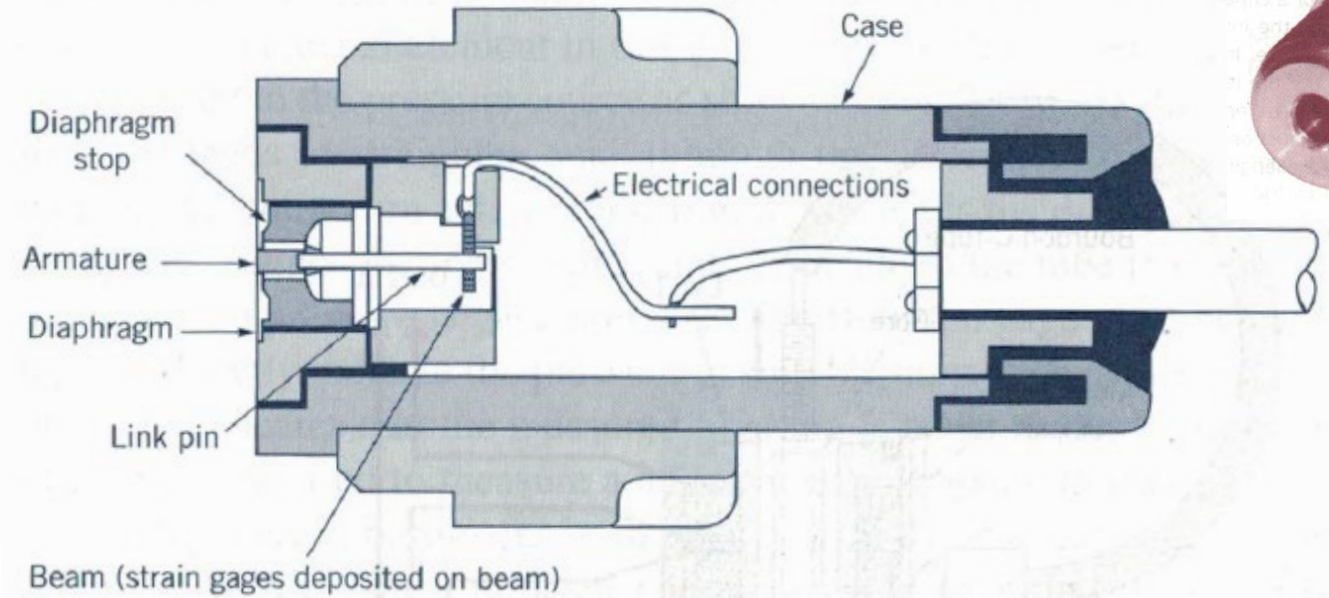
FIGURE 13.2 Bourdon tube pressure gage.

Catalogue of various gauges: https://www.omega.ca/en/sensors-and-sensing-equipment/pressure-and-strain/c/pressure-gauges

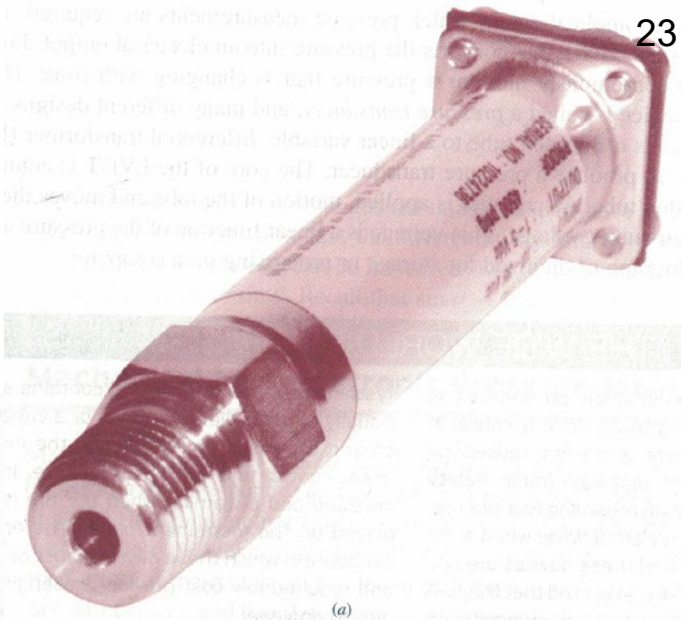
"pressure transducer" combines the sensor element of a gauge with a mechanical-to-electrical or mechanical-to-pneumatic converter and a power supply

Nice animation at https://www.futek.com/pressure-transducer

Catalogue of various transducers: https://www.omega.ca/en/sensors-and-sensing-equipment/pressure-and-strain/c/pressure-transducers



(b)



(a)

Figure 2.15 (*a*) **Photograph of a typical pressure transducer with a male thread fitting in front of the diaphragm for system connection and an electrical connector in the rear of the device.** (*b*) **Schematic diagram of a typical pressure transducer device (male thread connector not shown). Deflection of the diaphragm due to pressure is measured with a silicon beam on which strain gages and an associated bridge circuit have been deposited.**

Pressure Transmitters convert measured pressure (PV) to 4 to 20 mA signals

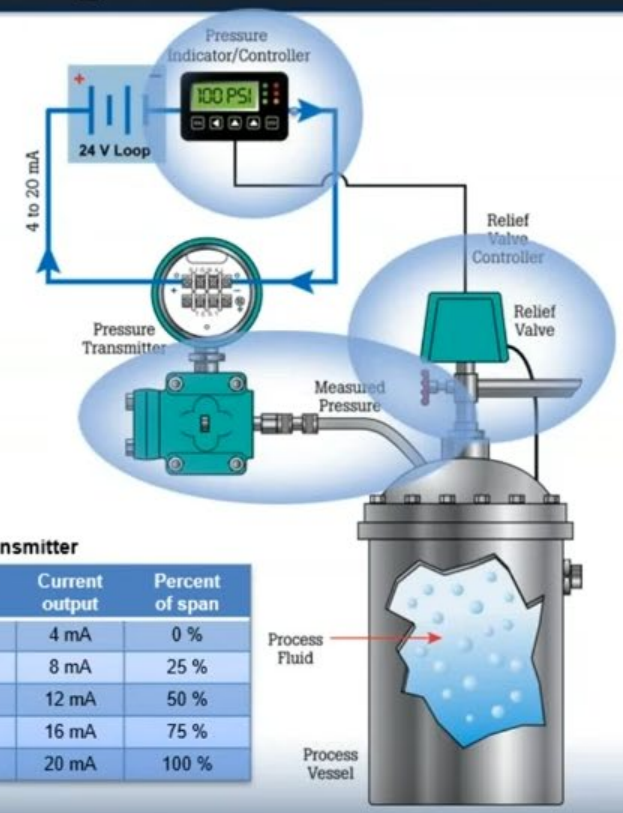
FLUKE

Measured PV:
The PV or primary/process variable in this example is the pressure being measured by the pressure transmitter.

Pressure Transmitter:
Senses the pressure of the process directly and converts the measurement to a 4 to 20 mA signal. The transmitter in this example has an input span (range) of 0 psi to 100 psi. The relationship of the measured pressure and mA signal is shown is the table.

Indicators/Controller:
In this example, the pressure indicator/controller is programmed to interpret a 4 mA signal as 0 psi and a 20 mA signal as 100 psi.

Pressure relief valve:
Final control element. In this example if the measured pressure is too high, the controller instructs the valve to open to reduce the pressure in the vessel.



Pressure transmitter

Pressure input	Current output	Percent of span
0 psi	4 mA	0 %
25 psi	8 mA	25 %
50 psi	12 mA	50 %
75 psi	16 mA	75 %
100 psi	20 mA	100 %

TRANSCAT®
Trust in every measure

9:09 / 50:58

https://www.youtube.com/watch?app=desktop&v=UYSrKKegQbc

Watch from 7:40 to 9:40

Differential Pressure

- Difference in pressure between two points
- Often used for flow measurements (see next section)
- Common equipment includes the following:

Manometer

Digital Manometer

Differential Pressure Gauge

Digital Differential Pressure Gauge

Designation: F2100 – 19

Standard Specification for Performance of Materials Used in Medical Face Masks[1]

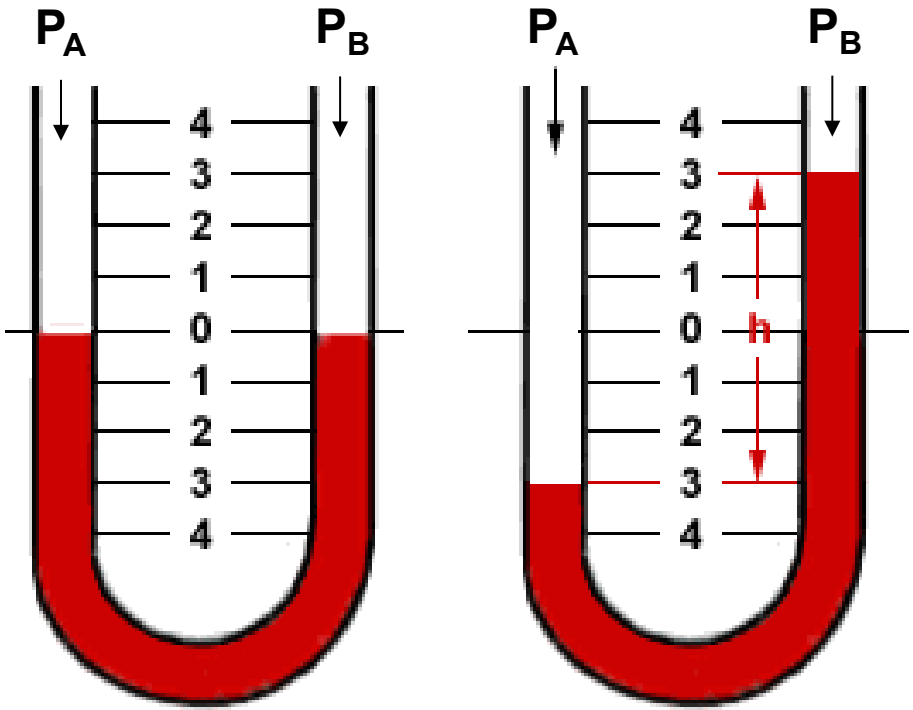
This standard is issued under the fixed designation F2100; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

TABLE 1 Medical Face Mask Material Requirements by Performance Level

Characteristic	Level 1 Barrier	Level 2 Barrier	Level 3 Barrier
Bacterial filtration efficiency, %	≥95	≥98	≥98
Differential pressure, mm H_2O/cm^2	<5.0	<6.0	<6.0
Sub-micron particulate filtration efficiency at 0.1 micron, %	≥95	≥98	≥98
Resistance to penetration by synthetic blood, minimum pressure in mm Hg for pass result	80	120	160
Flame spread	Class 1	Class 1	Class 1

Differential Pressure

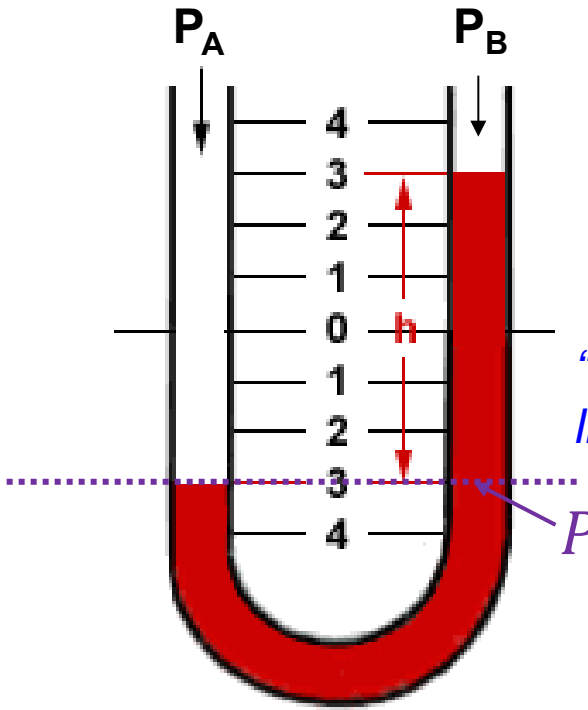
- Basic function of a manometer



$P_A = P_B$ $P_A > P_B$

Differential Pressure

- Basic function of a manometer



"Pressure increases linearly with fluid depth"

$$P_C = P_B + \rho g h$$

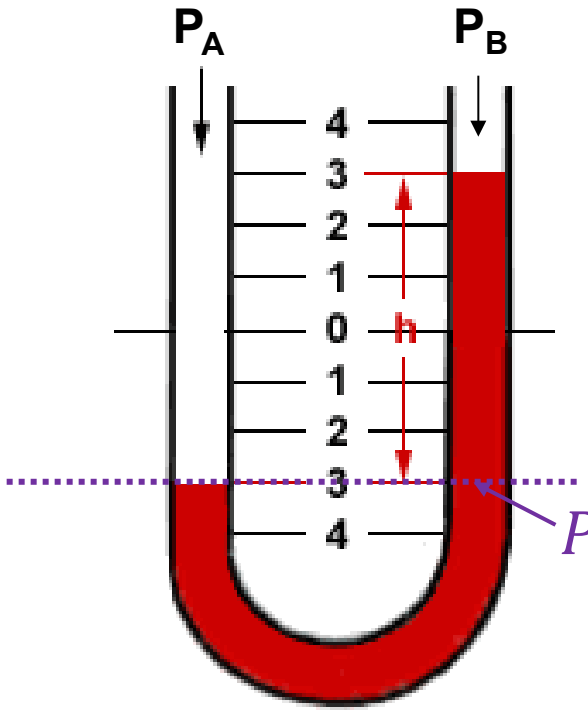
$P_A > P_B$

$$P_A = P_C$$

"the pressure is the same at all points on a horizontal plane in a given fluid provided that the points are interconnected by the same fluid!"

Differential Pressure

- Basic function of a manometer



$$P_C = P_B + \rho g h$$

$P_A > P_B$

$$P_A = P_C$$

$$P_A = P_B + \rho g h$$

Differential Pressure $P_A - P_B = \rho g h$

Appendix A – Property data

Table A.1

Specific Gravities of Selected Engineering Materials

(a) Common Manometer Liquids at 20°C

Liquid	Specific Gravity
E.V. Hill blue oil	0.797
Meriam red oil	0.827
Benzene	0.879
Dibutyl phthalate	1.04
Monochloronaphthalene	1.20
Carbon tetrachloride	1.595
Bromoethylbenzene (Meriam blue)	1.75
Tetrabromoethane	2.95
Mercury	13.55

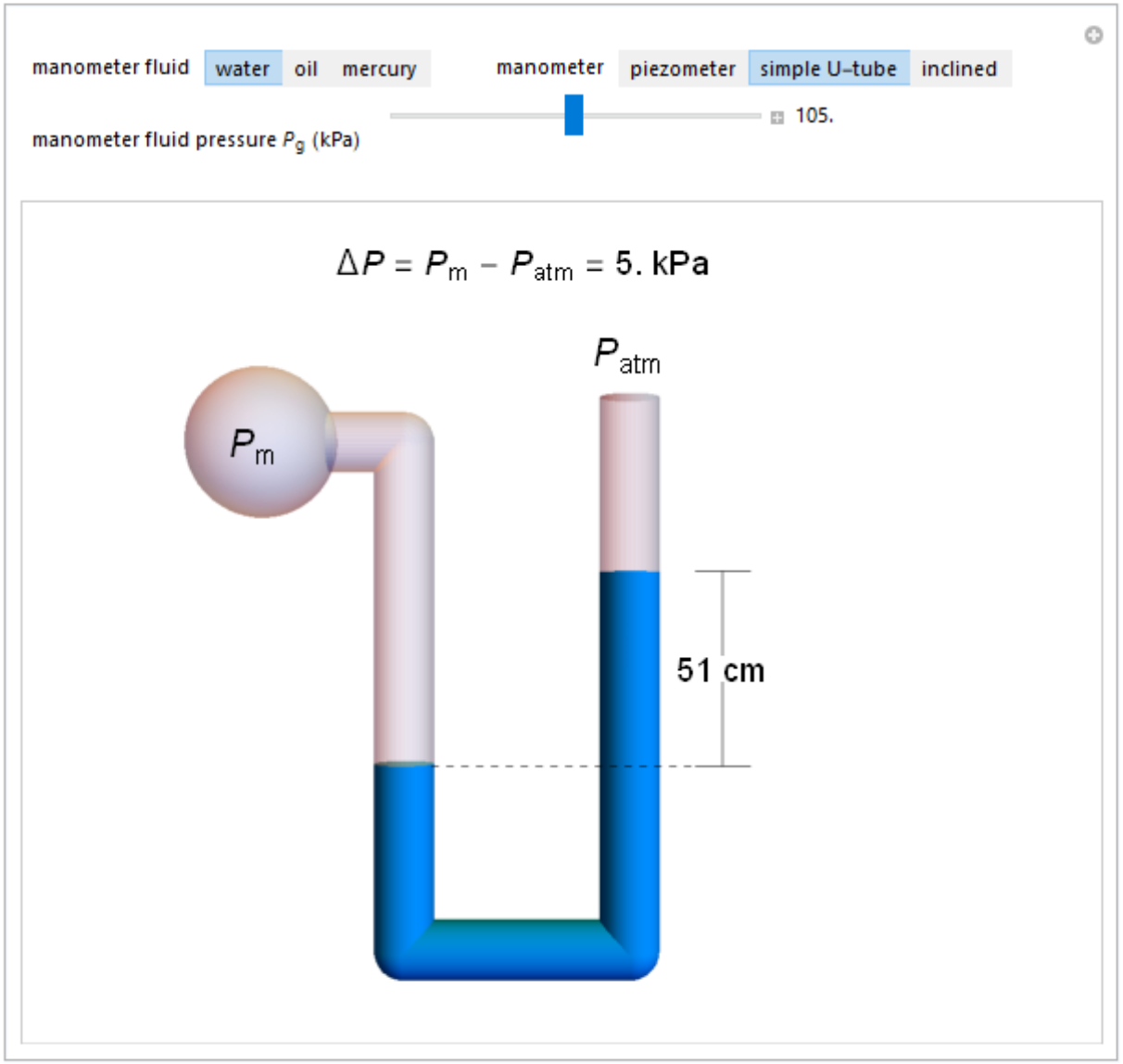
Source: Data from References [1–3].

Manometers - Wolfram CDF Player

File Edit Window Help

WOLFRAM CDF Player Find 100%

Manometers



Link to 'manometers' demo:

http://demonstrations.wolfram.com/Manometers/

Link to download CDF player:

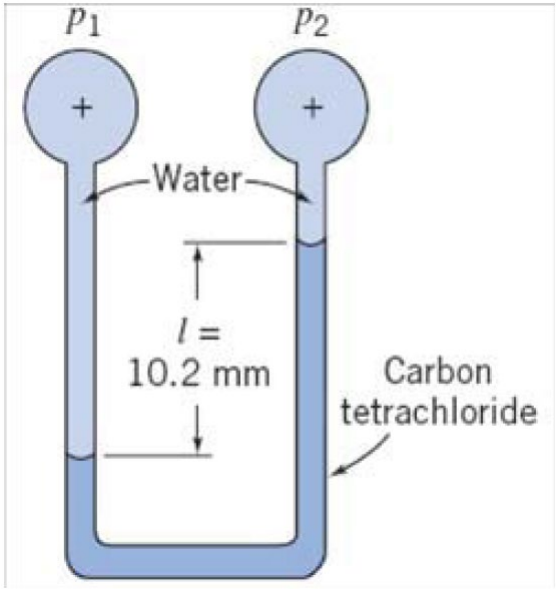
http://demonstrations.wolfram.com/download-cdf-player.html

In-class problem #1

3.11 (10th edition textbook)

3.17 (9th edition textbook)

Calculate the applied pressure difference for the two-fluid manometer shown in the figure to the right.



Answer:

$p_1 - p_2 = 59$ Pa

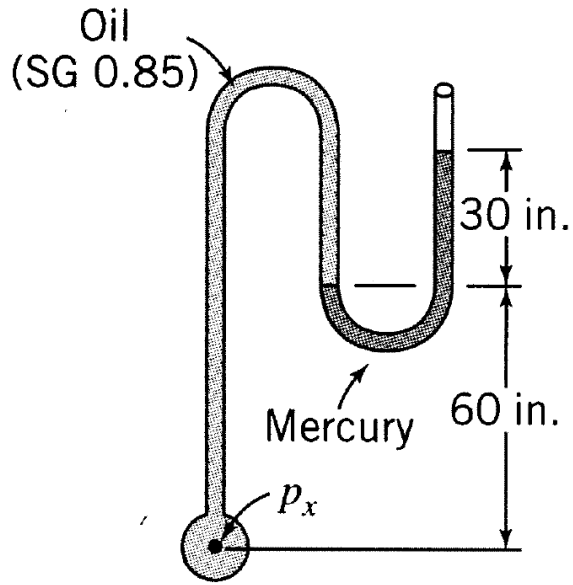
In-class problem #2

3.14 (10th edition textbook)

3.20 (9th edition textbook)

Determine the pressure p_x in the bulb for the manometer readings as shown in the figure.

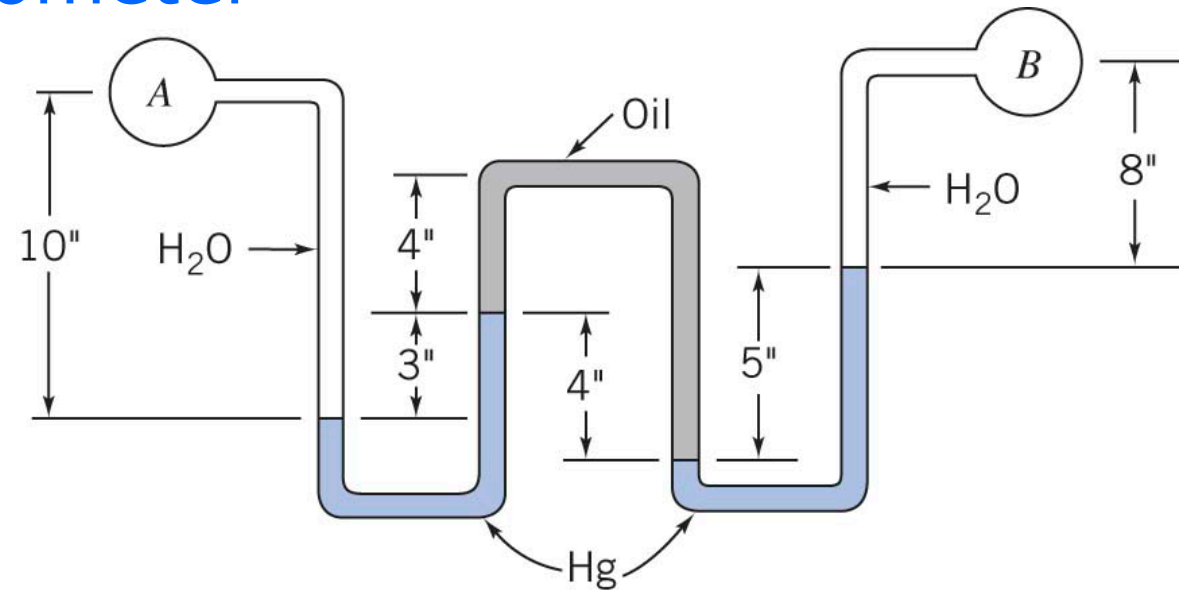
Answer:
p_x = 16.5 psi-g



P3.20

Multiple liquid manometer

For self-study, review Example 3.3



Flow Measurement

- "Almost all practical fluid engineering problems are associated with the need for an accurate flow measurement"
- Many different devices:

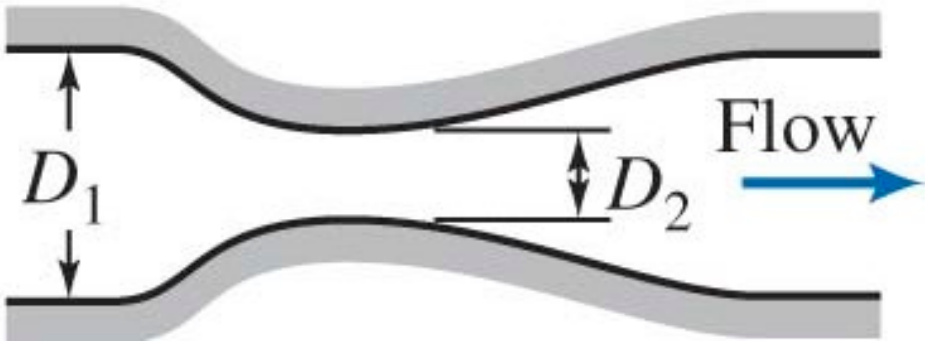
  "Knowledge of the underlying fluid mechanics and the physical principles involved are essential to selecting an appropriate measuring instrument"

  - Quantity meters – weigh/measure a volume of fluid over a known time increment
  - Indirect/rate meters:
    - Restriction flow meters
    - Linear flow meters
    - Other more specialized devices

ENCYCLOPEDIA OF CHEMICAL ENGINEERING EQUIPMENT

Restriction flow meters

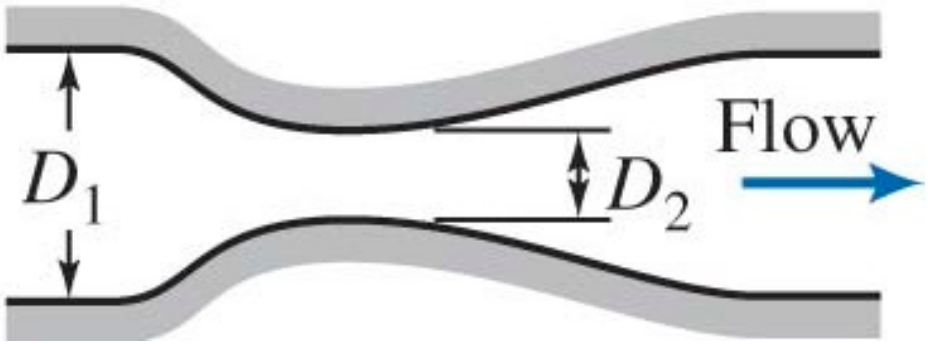
- Based on acceleration of a fluid stream through a nozzle of some shape:
  - Increase in fluid velocity ($V_2 > V_1$) causes decrease in pressure ($P_1 > P_2$)



$$\frac{p}{\rho} + \frac{V^2}{2} + gz = \text{constant} \qquad \textbf{\textit{Bernoulli Equation}} \qquad (6.8)$$

Restriction flow meters

- Based on acceleration of a fluid stream through a nozzle of some shape:
  - Increase in fluid velocity ($V_2 > V_1$) causes decrease in pressure ($P_1 > P_2$)



$$\frac{p}{\rho} + \frac{V^2}{2} + gz = \text{constant} \quad [=]\frac{\mathrm{m}^2}{\mathrm{s}^2}[=]\frac{\mathrm{kg}}{\mathrm{kg}}\frac{\mathrm{m}^2}{\mathrm{s}^2}[=]\frac{\mathrm{J}}{\mathrm{kg}} \tag{6.8}$$

flow energy per unit mass

kinetic energy per unit mass

potential energy per unit mass

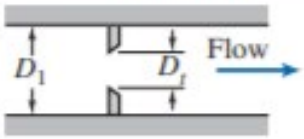
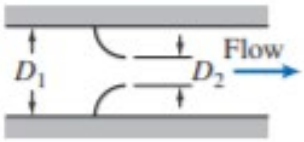
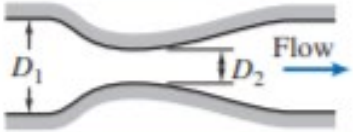
Restriction flow meters

- Based on acceleration of a fluid stream through a nozzle of some shape:
  - Increase in fluid velocity ($V_2 > V_1$) causes decrease in pressure ($P_1 > P_2$)

    Measure using pressure gauges or manometer
  - Infer flow rate (Q) using theoretical analysis and/or experimental correlations

Table 8.6

Characteristics of Orifice, Flow Nozzle, and Venturi Flow Meters

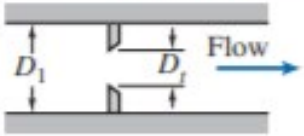
Flow Meter Type	Diagram	Head Loss	Initial Cost
Orifice		High	Low
Flow Nozzle		Intermediate	Intermediate
Venturi		Low	High

Restriction flow meters

- Based on acceleration of a fluid stream through a nozzle of some shape:
  - Increase in fluid velocity ($V_2 > V_1$) cause
  - Infer flow rate (Q) using theoretical anal correlations

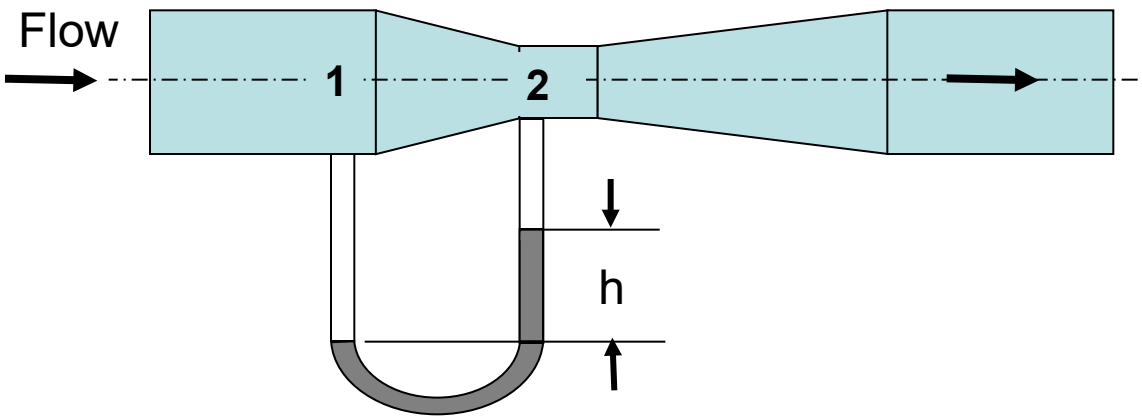
Table 8.6

Characteristics of Orifice, Flow Nozzle, and Venturi Fl

Flow Meter Type	Diagram
Orifice	

Venturi meter

- The conical diffuser section downstream from the throat gives excellent pressure recovery; thus overall energy loss is low.



- The difference in height between two legs of manometer (h) is proportional to pressure difference (P_1-P_2)
- Disadvantage is that the measured output (P_1-P_2) is not linear with flow rate (Q)!

'Throat' area, A_t

$$Q = \frac{\pi}{4} D_2^2 \sqrt{\frac{2 \cdot (P_1 - P_2)}{\rho \left(1 - \frac{D_2^4}{D_1^4}\right)}}$$

ρ is the density of the fluid in venturi

P_1 is the pressure 'upstream' of the venturi meter

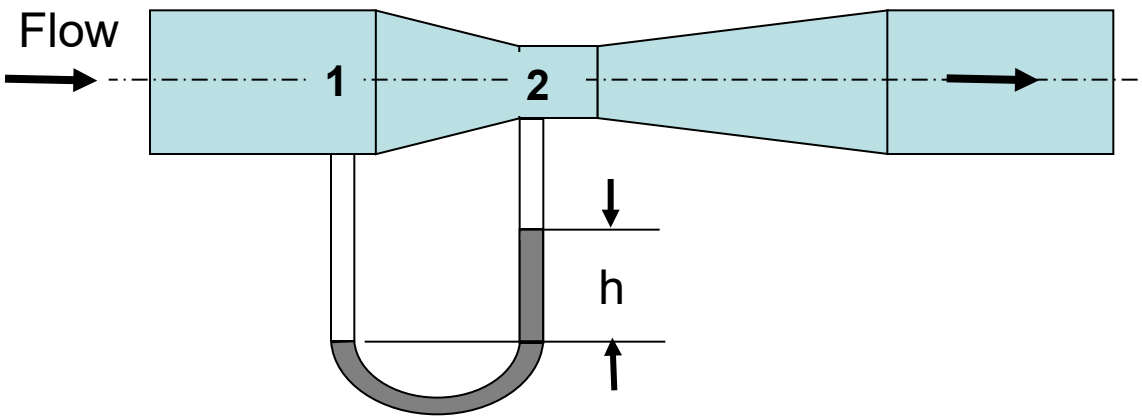
D_1 is the diameter 'upstream' of the venturi meter

P_2 is the pressure at the 'throat' of the venturi meter

D_2 is the diameter of the 'throat' in the venturi meter

Venturi meter

- The conical diffuser section downstream from the throat gives excellent pressure recovery; thus overall energy loss is low.



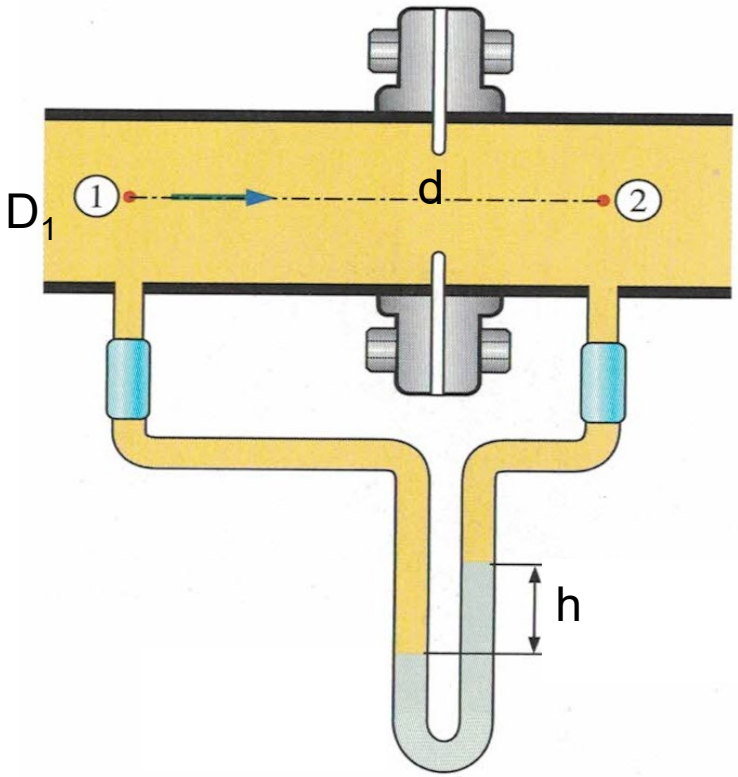
$$Q = \frac{\pi}{4} D_2^2 \sqrt{\frac{2 \cdot (P_1 - P_2)}{\rho \left(1 - \frac{D_2^4}{D_1^4}\right)}}$$

Derived from mass balance and mechanical energy balance (i.e. Bernoulli equation) – will come back to this later in the course!

- The difference in height between two legs of manometer (h) is proportional to pressure difference (P_1-P_2)
- Disadvantage is that the measured output (P_1-P_2) is not linear with flow rate (Q)!

Orifice meter

- The sudden change in flow area causes considerable swirl and thus significant energy loss; those losses are accounted for by incorporating a correction factor (the orifice coefficient, C_o) in the equation



$$Q = C_o \frac{\pi}{4} d^2 \sqrt{\frac{2 \cdot (P_1 - P_2)}{\rho \left(1 - \frac{d^4}{D_1^4}\right)}}$$

Close to Eqn. 8.54

ρ is the density of the fluid

P_1 is the pressure 'upstream' of the orifice meter

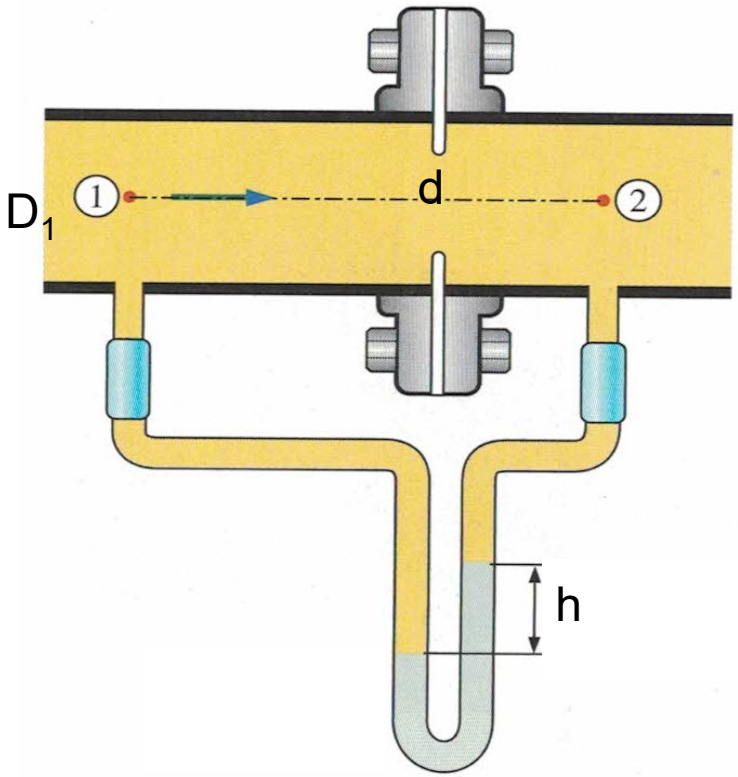
D_1 is the diameter 'upstream' of the orifice meter

P_2 is the pressure 'downstream' of the orifice meter

d is the diameter of the 'throat' in the orifice meter

Orifice meter

- The sudden change in flow area causes considerable swirl and thus significant energy loss; those losses are accounted for by incorporating a correction factor (the orifice coefficient, C_o) in the equation



$$Q = C_o \frac{\pi}{4} d^2 \sqrt{\frac{2 \cdot (P_1 - P_2)}{\rho \left(1 - \frac{d^4}{D_1^4}\right)}}$$

Close to Eqn. 8.54

β^4

'Throat' area, A_t

ρ is the density of the fluid

P_1 is the pressure 'upstream' of the orifice meter

D_1 is the diameter 'upstream' of the orifice meter

P_2 is the pressure 'downstream' of the orifice meter

d is the diameter of the 'throat' in the orifice meter

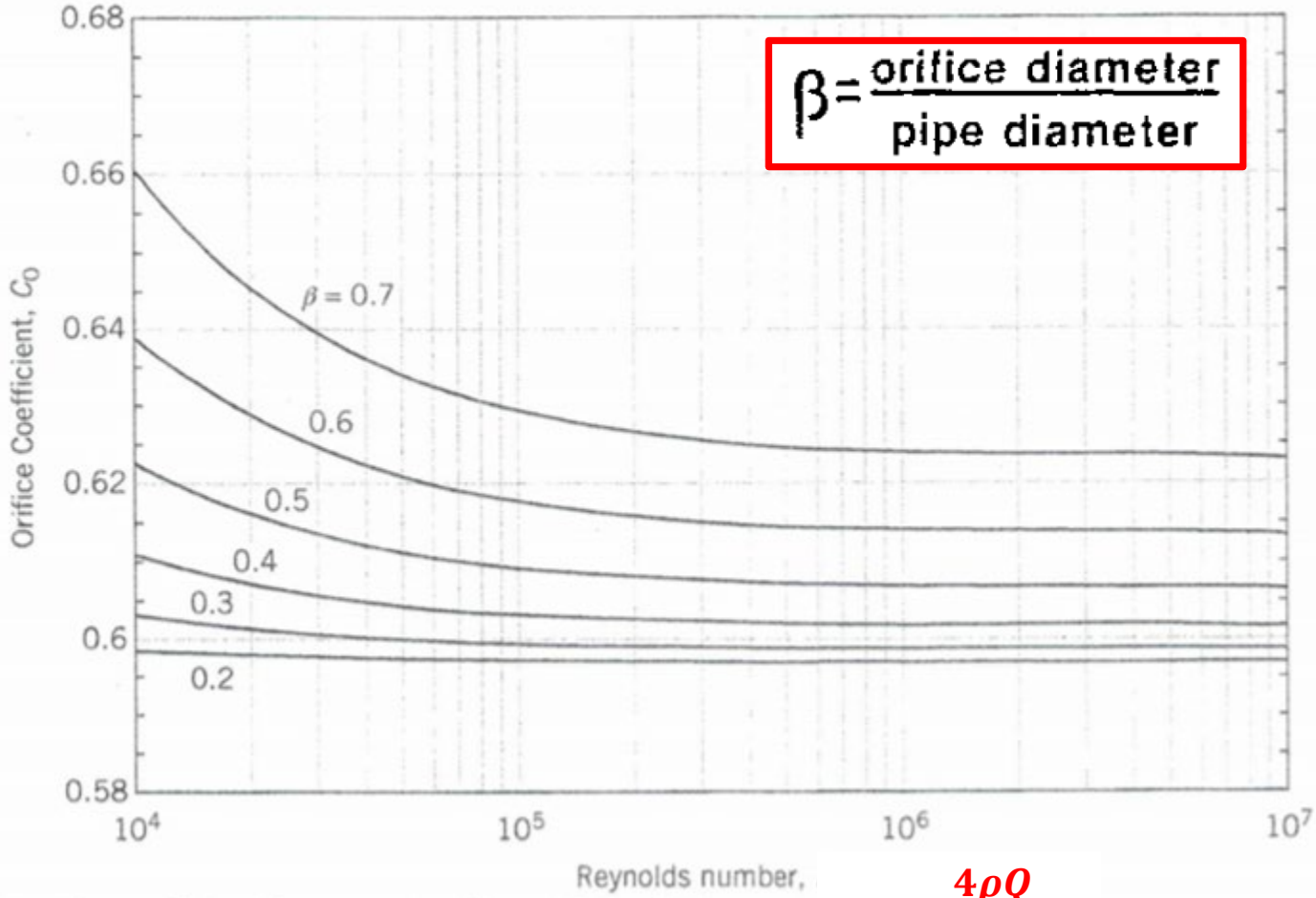


Fig. 8.20 Flow coefficient for concentric orifices with D and $D/2$ taps.

$$Re = \frac{4\rho Q}{\pi \mu D_1}$$

ρ is the density of the fluid

Q is the volumetric flow rate of the fluid

μ is the viscosity of the fluid

D_1 is the diameter 'upstream' of the orifice meter

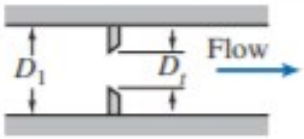
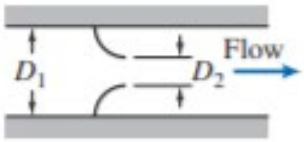
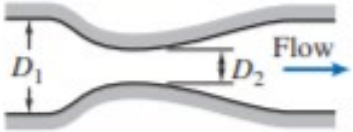
Restriction flow meters

- Based on acceleration of a fluid stream through a nozzle of some shape:
  - Increase in fluid velocity ($V_2 > V_1$) causes decrease in pressure ($P_1 > P_2$)

    Measure using pressure gauges or manometer
  - Infer flow rate (Q) using theoretical analysis and/or experimental correlations

Table 8.6

Characteristics of Orifice, Flow Nozzle, and Venturi Flow Meters

Flow Meter Type	Diagram	Head Loss	Initial Cost
Orifice		High C_o ~0.6	Low
Flow Nozzle		Intermediate	Intermediate
Venturi		Low C_o ~1	High

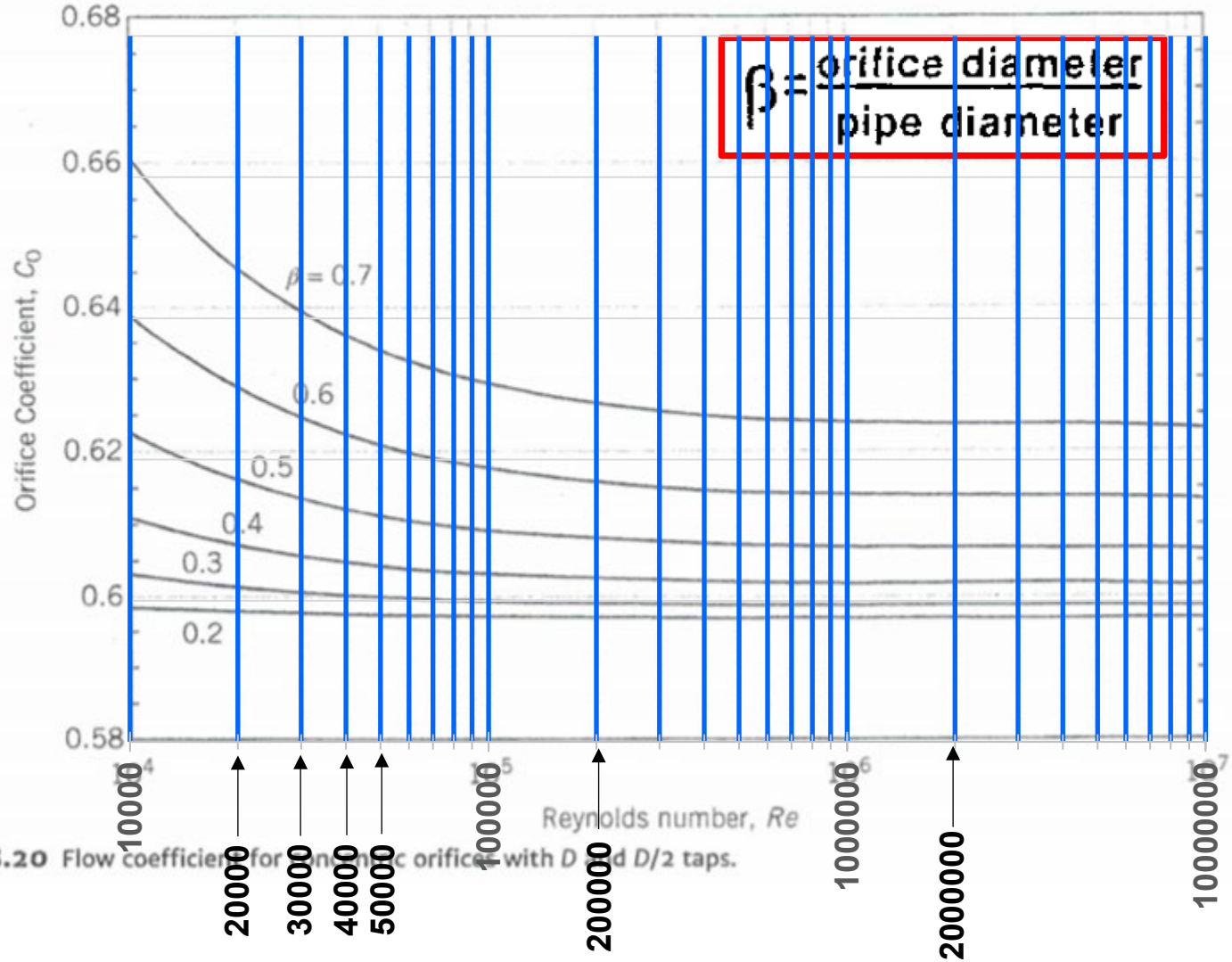


Fig. 8.20 Flow coefficient for concentric orifices with D and $D/2$ taps.

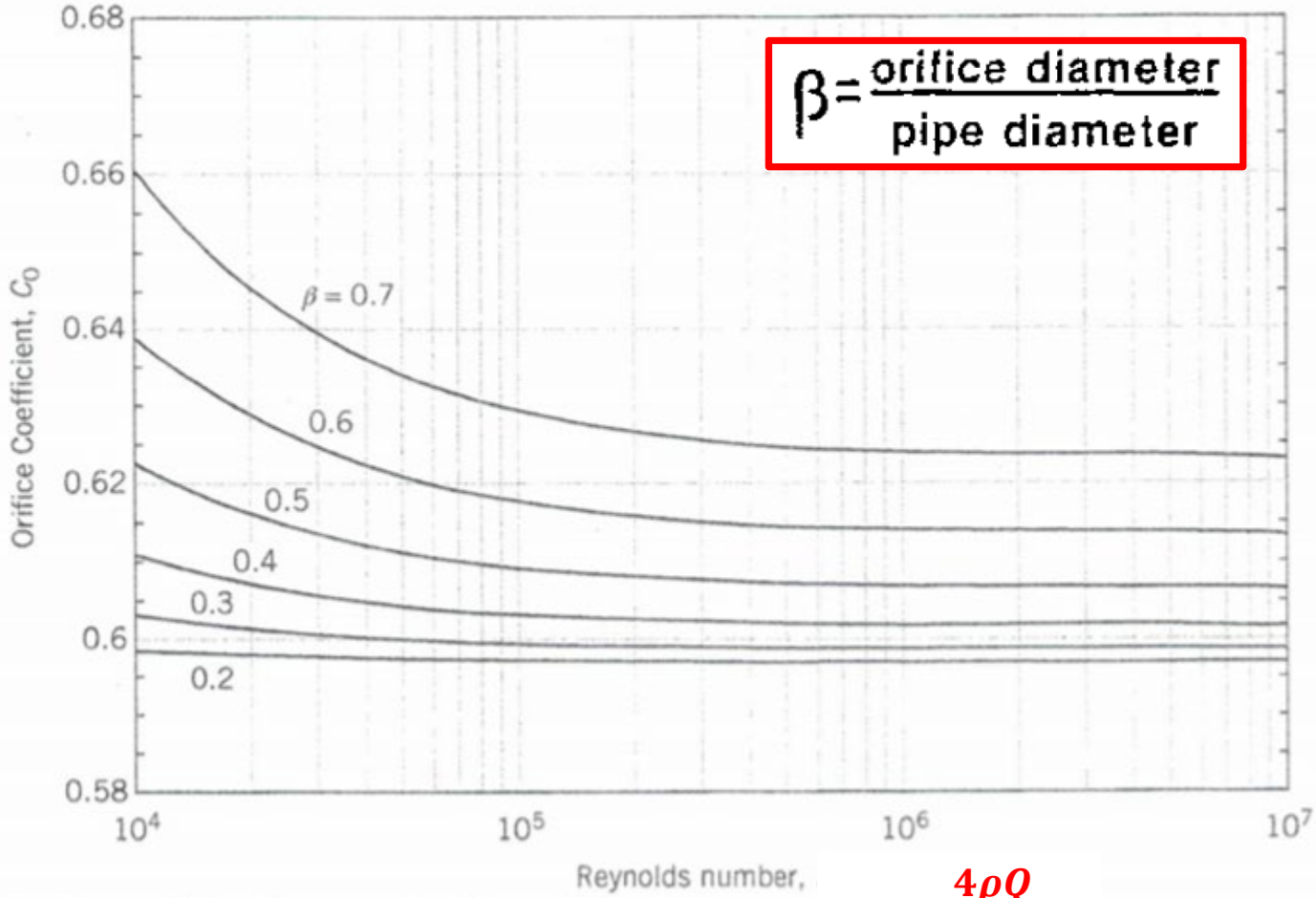


Fig. 8.20 Flow coefficient for concentric orifices with D and $D/2$ taps.

$$Re = \frac{4\rho Q}{\pi \mu D_1}$$

ρ is the density of the fluid

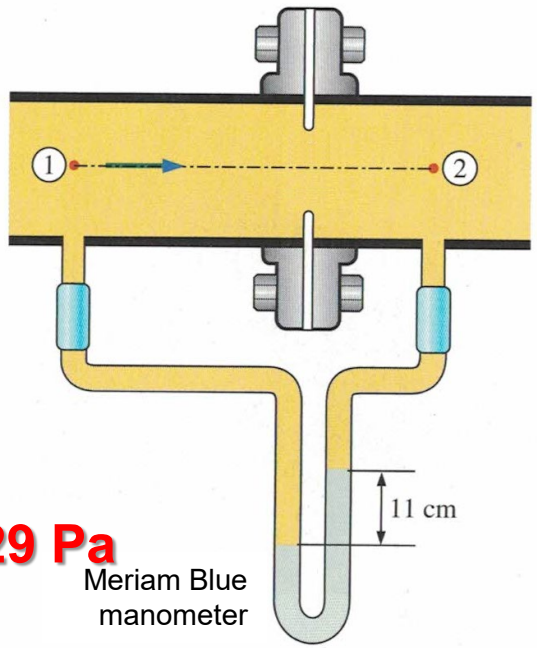
Q is the volumetric flow rate of the fluid

μ is the viscosity of the fluid

D_1 is the diameter 'upstream' of the orifice meter

Adapted from question on 2019 midterm #1:

The flow rate of methanol (at 20 °C) through a horizontal pipe (internal diameter = 4 cm) is to be measured using an orifice meter (internal 'throat' diameter = 2.8 cm) equipped with a Meriam Blue manometer as shown in the accompanying figure.



a. If the differential height of the two legs of the manometer is 11 cm, determine the pressure difference (in Pa) across the orifice meter (i.e. $P_1 - P_2$). **Ans = 1029 Pa**

b. Determine the flow rate (in L/s) of methanol through the horizontal pipe. **Ans = 0.73 L/s**

c. Determine the average velocity (in m/s) of methanol in the horizontal pipe where the internal diameter = 4 cm. **Ans = 0.58 m/s**

For self-study, review Example 8.12…but ignore part about "determine the head loss if the flow area at the…..".

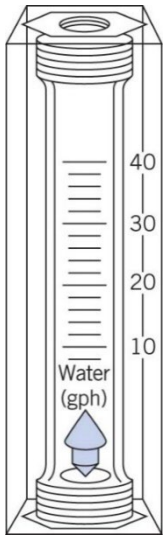
Linear flow meters

- No need to measure differential pressure

Rotameter

A vertical tapered tube through which the fluid flows and causes the “float” to move to a point where drag forces are balanced by weight and buoyancy forces

Tube is pre-calibrated with a flow rate scale



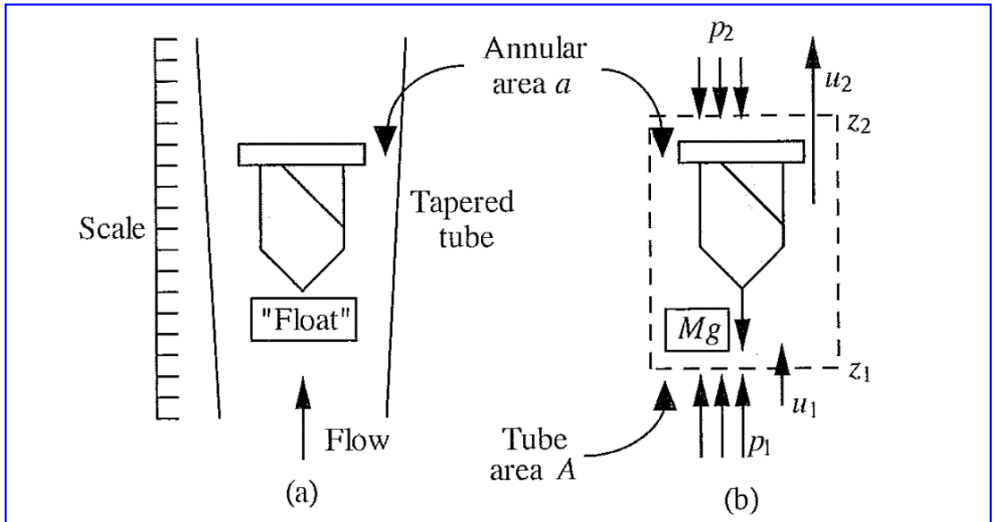


Fig. E2.6 (a) Section of rotameter, and (b) control volume.

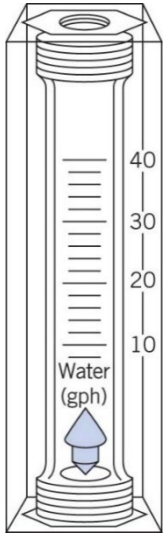
Linear flow meters

- No need to measure differential pressure

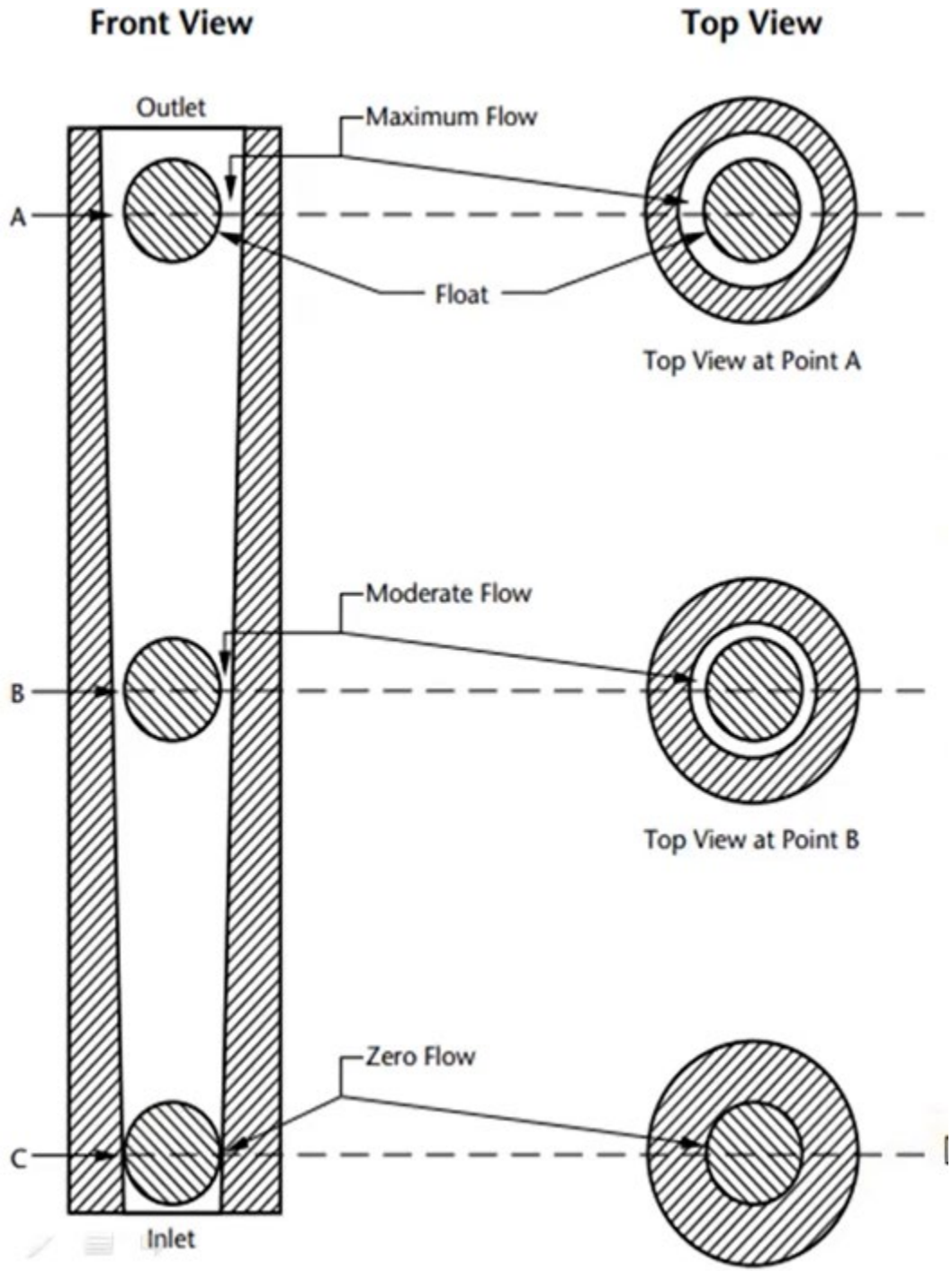
Rotameter

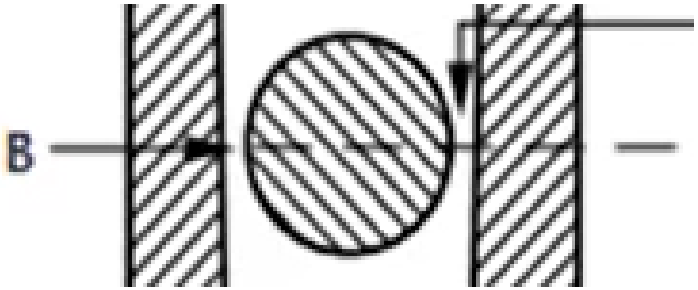
A vertical tapered tube through which the fluid flows and causes the “float” to move to a point where drag forces are balanced by weight and buoyancy forces

Tube is pre-calibrated with a flow rate scale



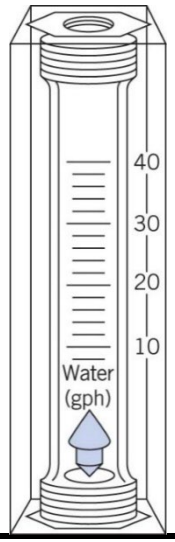
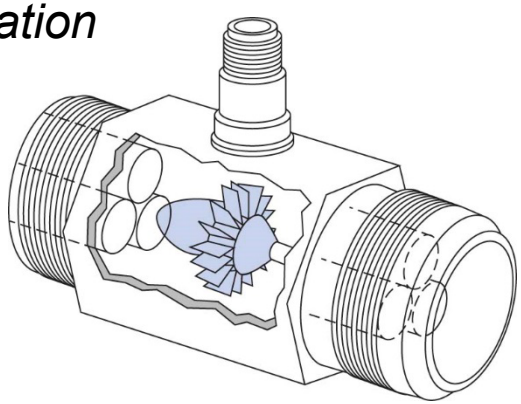
https://koboldusa.com/products/flow/variable-area-rotameters-flow-meters-switches/kfr-acrylic-flow-meter-liquids-gases/


Front View
Top View
Outlet
Maximum Flow
A
Float
Top View at Point A
Moderate Flow
B
Top View at Point B
Zero Flow
C
Inlet


B

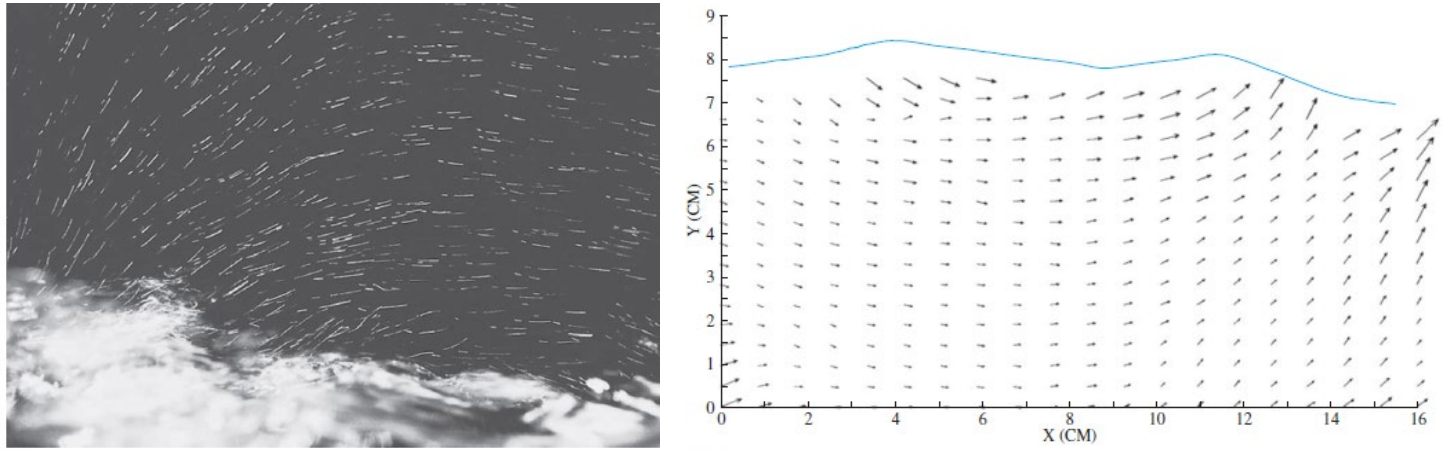
Linear flow meters

- No need to measure differential pressure

Rotameter	Turbine flow meter (aka propeller meter)
A vertical tapered tube through which the fluid flows and causes the “float” to move to a point where drag forces are balanced by weight and buoyancy forces Tube is pre-calibrated with a flow rate scale 	Flow rate is related to rotational speed of the impeller *Speed is measured by blade passing an electromagnetic pickup mounted in the casing* *Requires calibration* 

Specialized instruments/techniques

- Electromagnetic flowmeter
- Vortex-shedding meter
- Acoustic flowmeter
- Coriolis flowmeter
- Particle Image Velocimetry (PIV):
  - A pulsed light source illuminates small fluid-entrained particles (~1 – 20 μm) over short exposure times
  - A camera (synchronized with the light) records the location of the particles



(a) (b)

FIGURE 13.21 Particle Image Velocimetry (PIV): (a) photograph of particle pathlines; (b) scaled velocity vectors. (Courtesy of R. Bouwmeester.)

Problems from textbook for self-study

10th edition	9th Edition
3.11	3.17
3.12	3.18
3.13	3.19
3.16	3.25
3.19	3.23
8.51	n/a

PIPE-FLO

Tutorial 3 “Simple Fluid Statics”